BRAD BURCH

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SUMMARY

Software engineer with 7 years building and shipping production full-stack systems and LLM agents (Claude, MCP, OAuth 2.1 + PKCE), backed by a CI-gated Vitest + Playwright suite and automated deploys. Recent work runs on Cloudflare's edge stack (Workers, D1, KV, R2) with TypeScript on Hono and React 19. Owns systems from schema design through ship and on-call.

SKILLS

Languages: TypeScript, JavaScript, Python, Go, Java, Scala, SQL
Frameworks & Runtimes: React 19, React Router 7, Node.js, Hono, Spring, Play, Cloudflare Workers
Infrastructure: Cloudflare (Workers, D1, KV, R2), AWS (DynamoDB, SQS), PostgreSQL, MSSQL, SQLite, Docker
AI / LLM: Claude (Haiku, Sonnet), Vercel AI SDK, Model Context Protocol (MCP), agentic tool loops, streaming SSE, AI-assisted development (Claude Code, Cursor, AI coding agents)
Auth & Security: OAuth 2.0, PKCE (RFC 7636), JWT (HS256), SAML 2.0, refresh token rotation, CSRF and XSS hardening
Testing & Delivery: Vitest, Playwright, GitHub Actions, CI/CD, Jenkins, structured JSON logging, production monitoring & alerting

EXPERIENCE

Founder & Software Engineer — Sept 2022 - Present
Brad Paws, San Francisco, CA

- Designed and built a production TypeScript monorepo (React Router 7 client portal, Hono API Worker, scheduled sync Worker, shared domain package) on Cloudflare Workers, D1, KV, R2, and Workflows, backed by a 23-table D1 schema and no origin server; it runs a live business with 30+ active clients, 60+ bookings/month, and 30+ invoices/month.
- Built OAuth 2.1 + PKCE (RFC 7636) from scratch for an MCP server, then hardened the app against the attack classes a payments-adjacent product faces: revocable server-side sessions (a SessionVersion checked fail-closed on every request), SPF/DKIM-verified payment ingestion, and RFC 5987 filename encoding that closes a header-injection vector.
- Shipped a production AI chat agent (Claude Haiku 4.5, Vercel AI SDK, 15 tools) with two-phase confirmation on mutations and layered cost controls: per-user rate limits, daily token budgets, a circuit breaker, and hard per-user and global USD caps via a reserve-then-reconcile scheme.
- Modeled the core booking domain: conflict-aware scheduling across three service types with pet-count capacity rules (not event counts), a server-enforced cancellation policy with tiered fees anchored to the Pacific business day, and combined-set multi-pet pricing that keeps every quote, calendar charge, and invoice in agreement.
- Re-architected a fragile calendar backfill into a Cloudflare Workflows fan-out/fan-in with exactly-once finalization, and built an automated test suite (Vitest + Playwright, CI-gated) audited for assertion quality rather than line coverage; the audit surfaced and closed four silently-untested security controls.
- Extracted two components from the production system and open-sourced them: Pawservation, an embeddable multi-tenant booking widget on Cloudflare Workers, and mcp-auth-kit, an OAuth 2.1 + PKCE MCP server kit (github.com/bradburch).

Software Engineer — May 2022 - Aug 2022
Front, San Francisco, CA (role eliminated in company-wide layoff)

- Shipped a Node.js REST API enabling account-level email rules for multi-domain enterprise customers, previously configurable only per-inbox.
- Performed a security upgrade on the core JavaScript email library responsible for 200K+ outbound messages per month.

Software Engineer — May 2019 - Aug 2021
Total Brain, San Francisco, CA

- Shipped a Scala backend integration for Apple, Google, and Facebook authentication, contributing to a 60% lift in sign-ups.
- Migrated user data from MSSQL to DynamoDB with a redesigned NoSQL schema, cutting API response times by 50%.
- Built an in-memory caching layer in front of DynamoDB to relieve hot read paths; standardized SAML 2.0 and JWT for enterprise SSO.
- Built a Jenkins CI/CD pipeline and bash-based release automation, increasing deploy frequency and adding structured logs for production observability.

Software Engineer — Jul 2017 - Aug 2018
Salesforce.com, San Francisco, CA

- Built automated testing frameworks on a developer-productivity team, including a persistent test-state repository in Java, PostgreSQL, and EclipseLink.
- Built an OAuth2 security filter for microservice API access control with remote identity validation.
- Built a React management UI that internal engineers used to configure, launch, and inspect microservice test runs, cutting internal test setup time by 40%.
- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams to ship on schedule.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science — 2017